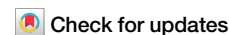




Observationally constrained global warming hysteresis under CO₂ removal



Se-Yong Song¹, Duo Chan²✉ & Sang-Wook Yeh³✉

Stabilizing global warming under ongoing anthropogenic warming will likely require atmospheric carbon dioxide removal. However, the extent to which carbon dioxide removal can reverse past warming remains uncertain due to divergent responses across a limited set of models. Here, we develop a data-driven emulator based on an energy balance model with three layers that captures the nonlinear temperature response to forcing. Calibrated to state-of-the-art instrumental temperature datasets, our observationally constrained emulator projects that, at the same carbon dioxide concentration, global warming during the carbon dioxide removal phase will be on average 0.7 °C (95% confidence interval: [0.5, 1.0]) warmer than during the carbon dioxide increasing phase, narrowing the 0.2–1.9 °C inter-model range by more than 70%. Applied to mitigation scenarios, global warming is projected to temporarily overshoot the 1.5 °C target, even under stringent pathways reaching net-zero carbon dioxide emissions.

Despite ongoing efforts to reduce CO₂ emissions, atmospheric CO₂ levels continue to increase, with global warming reaching the 1.5 °C threshold in the recent observational record^{1–3}. Given this trajectory, achieving international climate targets increasingly relies on carbon dioxide removal (CDR) to counteract excess emissions and stabilize global temperatures^{4–6}. However, the effectiveness of CDR in reversing past warming remains uncertain, as the potential hysteresis effects in the climate system may delay or even limit the recovery^{7,8}.

Advances in modelling capability, including the CDR Model Intercomparison Project (CDRMIP)⁹, have provided insights into the susceptibility of hysteresis and irreversibility across various climate system components^{10–13}. Climate models suggest that even if CO₂ concentrations are restored to pre-industrial levels, various climate states, including global mean surface temperature (GMST), the Antarctic ice sheet, sea-level rise, ocean acidification, the Atlantic Meridional Overturning Circulation (AMOC), and hydrological cycles, may not fully recover to their previous state^{14–19}. As such, hysteresis denotes the path dependence of the climate response during periods of CO₂ increase and decrease, while reversibility refers to the ability of the climate system to return to its initial state. This effect manifests on different recovery timescales and across regions, stemming from inertia effects in individual climate system components^{20,21}.

Despite insights from Earth System Model (ESM) simulations, in particular, the projected magnitude of GMST hysteresis remains substantially uncertain across models^{22–24}. Such uncertainty is concerning

because some models fail to reproduce a reasonable historical warming²⁵. Moreover, the limited sets of model simulations used in the CDRMIP⁹ may be insufficient for properly quantifying uncertainties or guiding meaningful emergent constraints^{23,26}. These knowledge gaps clearly challenge targeted long-term policy making and climate action designs^{27,28}.

To understand diverging ESM projections, theories based on energy balance models (EBMs) have been proposed^{29,30}. These models, often containing only two layers representing the surface and interior ocean³¹, are used to reproduce ESM simulations and explain hysteresis mechanisms using processes such as climate feedback strength and ocean heat uptake efficiency^{32,33}. Although such EBMs have been used to constrain 21st-century warming projections³⁴, their reproducibility under mitigation pathways including CDR and zero-emission scenarios remains limited, as key aspects of the non-linear temperature response during the recovery phase are not adequately captured^{29,30}. This lack of consensus, along with the limited sets of simulations, hinders our ability to constrain GMST hysteresis. As a result, inter-model differences in hysteresis response under CDR scenarios have yet to be properly constrained, leading to low confidence in projections of long-term climate reversibility³⁵.

Here, we aim to provide a constrained estimate of GMST hysteresis under the CDR-reversibility scenario with comprehensive uncertainty quantification. Specifically, we propose a three-layer EBM that explicitly incorporates the non-linear response to external forcing, thereby effectively emulating the global temperature recovery trajectories under the CDR scenario as simulated by ESMs. Uncertainties accounted for include

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parameter uncertainties via Bayesian inferences, errors in observational constraints, and EBM's structural uncertainty.

Results

Improving the representation of GMST hysteresis using a three-layer EBM

The CDR scenario targeted here follows a symmetric CO_2 forcing trajectory, in which atmospheric CO_2 concentration increases 284.7 ppm to 1138 ppm at a rate of 1% per year during the ramp-up phase, followed by a decline back to 284.7 ppm during the ramp-down phase. The degree of hysteresis is defined using time windows symmetric about the CO_2 peak, with our primary analysis focusing on long-term CDR effect after CO_2 peaks estimated from the temperature difference between the ramp-up (1–139 years) and ramp-down (141–279 years) periods on the same atmospheric CO_2 level^{20,23}. It is worth noting that the degree of hysteresis depends on both the CO_2 ramp rate and the choice of the window length²⁰. Here, we explicitly scope our analysis under the standard CDRMIP 1% CO_2 ramp rate CDR-reversibility protocol⁹. We further test the sensitivity of the hysteresis definition to the choice of window length, but our main conclusions remain robust (see “Methods” section), with the implications discussed in a later section.

All eight ESMs participating in the CMIP6 CDRMIP experiment show delayed recovery in the GMST response under the CDR scenario (Supplementary Fig. 1a). However, the degree of hysteresis in the GMST response, quantified as the average temperature difference between the ramp-up (1–139 years) and ramp-down (141–279 years) trajectories, conditions on the same atmospheric CO_2 level, diverges substantially across ESMs

(Supplementary Fig. 1b). Such a large spread points to the substantial uncertainty in ESM projections of global temperature responses to CDR pathways²³.

A two-layer EBM (hereafter EBM2; see Fig. 1a for a schematic) is used to emulate the GMST response under the CDR scenario (see “Methods” section). Using Bayesian inference, we calibrate the EBM2 parameters based on the abrupt-4x CO_2 simulation (Fig. 1a for an example of CESM2 and Supplementary Fig. 2 for all models), the experiment widely used to diagnose ESM behaviour^{36,37}. The resulting posterior distributions of the EBM2 parameters (Supplementary Table 1) are then used to generate global temperature responses under the CDR scenario (see “Methods” section). Compared to each ESM simulation, EBM2 produces a systematically warmer GMST during the ramp-down period (Fig. 1b for an example of CESM2 and Supplementary Fig. 3 for all models). This leads to an overestimated GMST hysteresis by about 49% on average and up to 262% for NorESM2-LM (Supplementary Fig. 1b). It suggests that EBM2 may not adequately capture the non-linear GMST response to CDR, as it assumes a linear global energy imbalance response to warming when fitting EBM2 parameters (Fig. 1a). Moreover, this mismatch indicates the limited utility of EBM2 for predicting long-term GMST evolution under the CDR scenario.

To better account for the non-linear behaviour, we explicitly allow EBM to change the climate feedback parameter (λ) if warming exceeds certain thresholds. The state dependence of λ may arise from radiative feedbacks that depend on evolving SST patterns, often referred to as the pattern effect^{38,39}. Taking CESM2 as an example, the λ decreases by $0.66 \text{ W m}^{-2} \text{ } ^\circ\text{C}^{-1}$, from 1.12 to $0.47 \text{ W m}^{-2} \text{ } ^\circ\text{C}^{-1}$, at a warming level of $4.1 \text{ } ^\circ\text{C}$ under the abrupt-4x CO_2 scenario (Supplementary Fig. 4), as indicated by

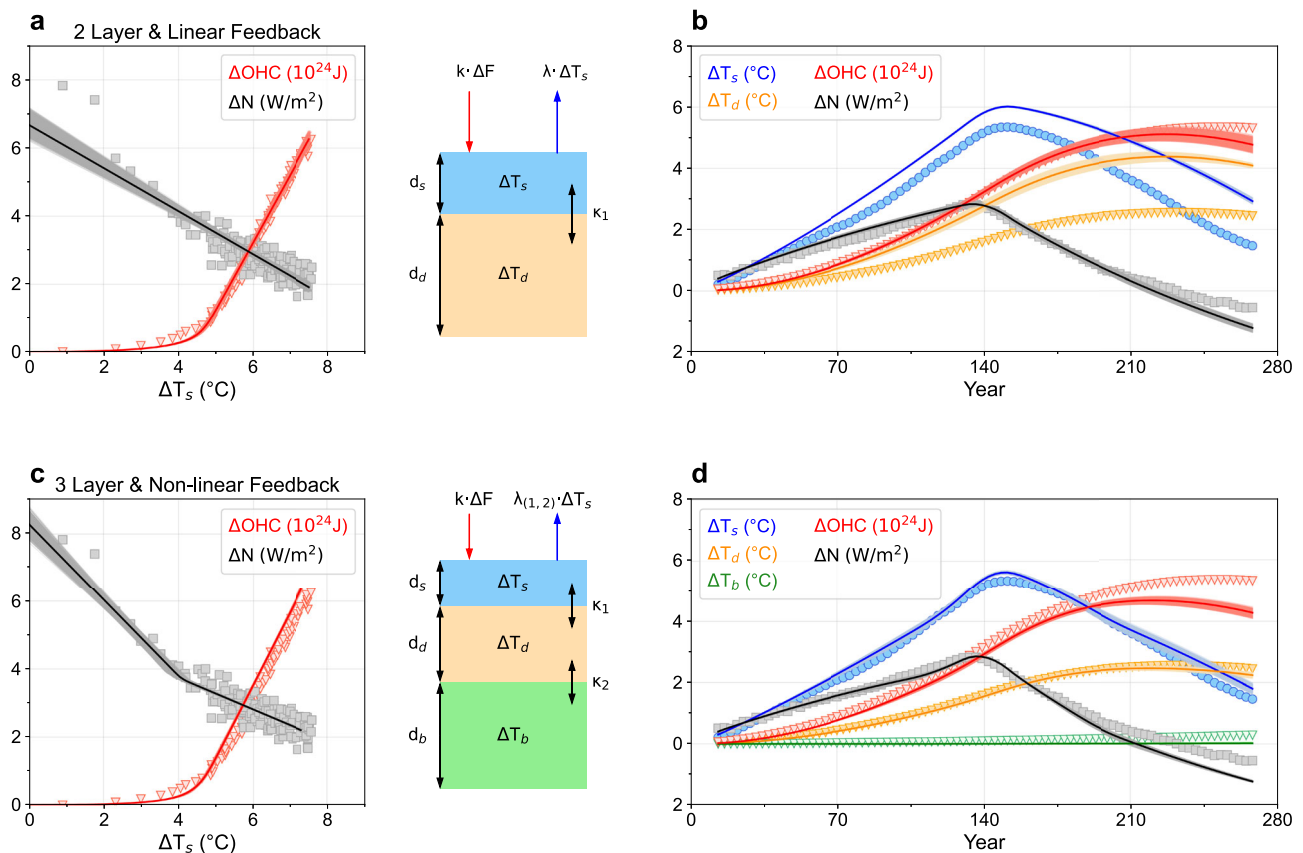
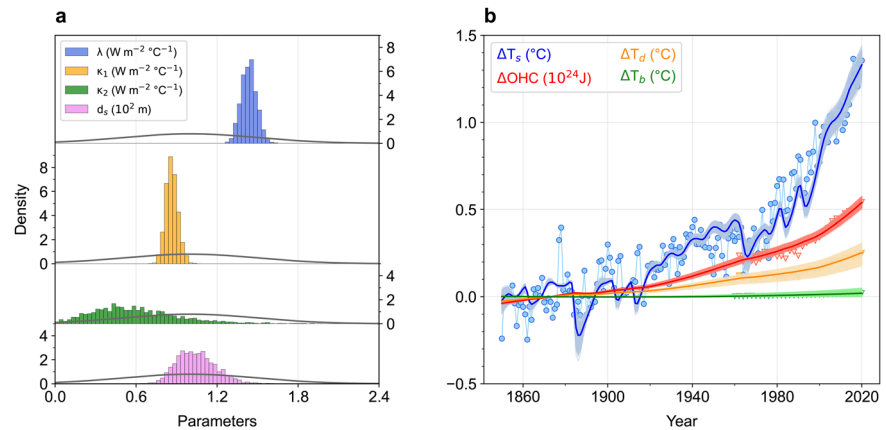


Fig. 1 | Global warming hysteresis emulated by two- and three-layer EBM. **a**, **c** Scatter plot of global energy imbalance (N) and ocean heat content (OHC) anomalies versus global mean surface temperature (T_s) anomaly from CESM2 under the abrupt-4x CO_2 scenario, with schematic diagrams of the two-layer (EBM2; **a**) and three-layer (EBM3; **c**) EBMs under linear and non-linear climate feedback frameworks, respectively (see “Methods” section for a detailed description of the

EBM). **b** Time series of the 21-year running mean of N , T_s , deep ocean temperature (T_d), and OHC anomalies from CESM2 (markers) and EBM2 emulation (lines) under the CDR scenario. **d** Same as (**b**), but including bottom-layer ocean temperature (T_b) from EBM3 emulation. Thin lines show individual EBM ensemble members, and thick lines denote their ensemble median.

Fig. 2 | Historical global warming emulated by EBM3 calibrated using observational estimates.

a Posterior probability density distributions of EBM3 parameters calibrated using the DCENT and IAP datasets: climate feedback parameter (λ), ocean heat uptake coefficients (κ_1 and κ_2), and the effective depth of the surface layer (d_s). Grey lines indicate the normal priors: $N(1, 0.5)$ for λ and κ , and $N(100, 50)$ for d_s . **b** Time series of T_s , T_d , T_b , and OHC anomalies from DCENT and IAP datasets (lines with markers) and from EBM3 emulations (lines). Thin lines show individual EBM ensemble members, and thick lines denote their ensemble median.



the median values of the 10,000 posterior samples inferred through Bayesian inference (see “Methods” section). An alternative approach is to assume that non-linearity occurs with a fixed time scale rather than critical temperature thresholds, e.g., in ref. 40. However, investigating CESM2 under the abrupt-2 $\times$ CO₂ scenario reveals that its temperature does not exhibit a non-linear response where warming does not exceed 4.1 °C (Supplementary Fig. 4). Such difference between abrupt-4 $\times$ CO₂ and abrupt-2 $\times$ CO₂ simulations have been reported to arise from non-linear changes in effective radiative forcing, heat capacity, and climate feedbacks that do not scale linearly with CO₂ forcing levels^{41–43}.

Another reason for the mismatch between EBM2 and ESM simulations could involve the vertical structure of the interior ocean’s temperature evolution. At greater depths, the ocean temperature response to radiative forcing becomes more delayed and recovers more slowly⁴⁴ (Supplementary Fig. 5). In response to CDR, the upper 100 m, primarily corresponding to the mixed layer, warms and recovers rapidly, whereas the layer below—extending to the upper 700 m near the climatological global pycnocline depth—adjusts more slowly with a gradual recovery. Below this layer, the ocean continues to warm while accumulating anthropogenic heat. In other words, the simplified two-layer ocean representation in EBM2 may limit its ability to adequately emulate the direction of heat redistribution across the water column under CDR (Supplementary Fig. 5), with implications for surface temperature responses⁴⁵. It, therefore, manifests as at least three distinct depth-dependent layers under the CDR scenario, although the depth boundaries vary slightly between models.

Motivated by these limitations, here we employ a three-layer EBM (hereafter EBM3; see Fig. 1c for a schematic) to explicitly represent the time-varying λ ^{38,39} and distinct timescales in the ocean interior’s temperature response^{45–47}. Using Bayesian inference, we explicitly allow λ to change as a function of a critical warming threshold (see “Methods” section; Fig. 1c for an example of CESM2 and Supplementary Fig. 2 for all models), yielding posterior samples for each EBM3 parameter under the abrupt-4 $\times$ CO₂ simulation (Supplementary Table 2). These samples are then used to emulate global temperature responses under the CDR scenario (see “Methods” section).

EBM3 shows improved agreement in reproducing the temperature evolution under the CDR scenario, with a notably reduced systematic warm bias during the ramp-down period (Fig. 1d for an example of CESM2 and Supplementary Fig. 3 for all models). When evaluated against ESM, EBM3 has a lower root-mean-square error (RMSE) than EBM2 in emulating the surface and deep ocean temperature response to CDR, except for the surface temperature in UKESM1-0-LL (Supplementary Table 3). As a result, EBM3 better captures GMST hysteresis, reducing the averaged difference to 21% compared to 49% in EBM2 relative to the ESM (Supplementary Fig. 1b).

It is worth noting that the EBM parameters are fitted to the abrupt-4 $\times$ CO₂ simulation and then evaluated using the CDR scenario. In contrast to calibrating and evaluating on the same CDR simulations, this out-of-

sample approach reduces concerns about CDR-specific overfitting arising from EBM3’s additional parameters. Instead, the improved fidelity of EBM3 in reproducing CDR-induced temperature evolution supports the interpretation of structural improvements in capturing the different timescales of temperature response under the assumption of a non-linear radiative response to warming that EBM2 cannot account for. This is further supported by the fact that EBM3 exhibits increasingly lower RMSE than EBM2 in GMST hysteresis as the window length used to define hysteresis becomes longer (see “Methods” section; Supplementary Fig. 1c). Thus, the enhanced performance of EBM3 is not merely a statistical artefact but a robust and consistent feature across ESMs (Supplementary Fig. 3).

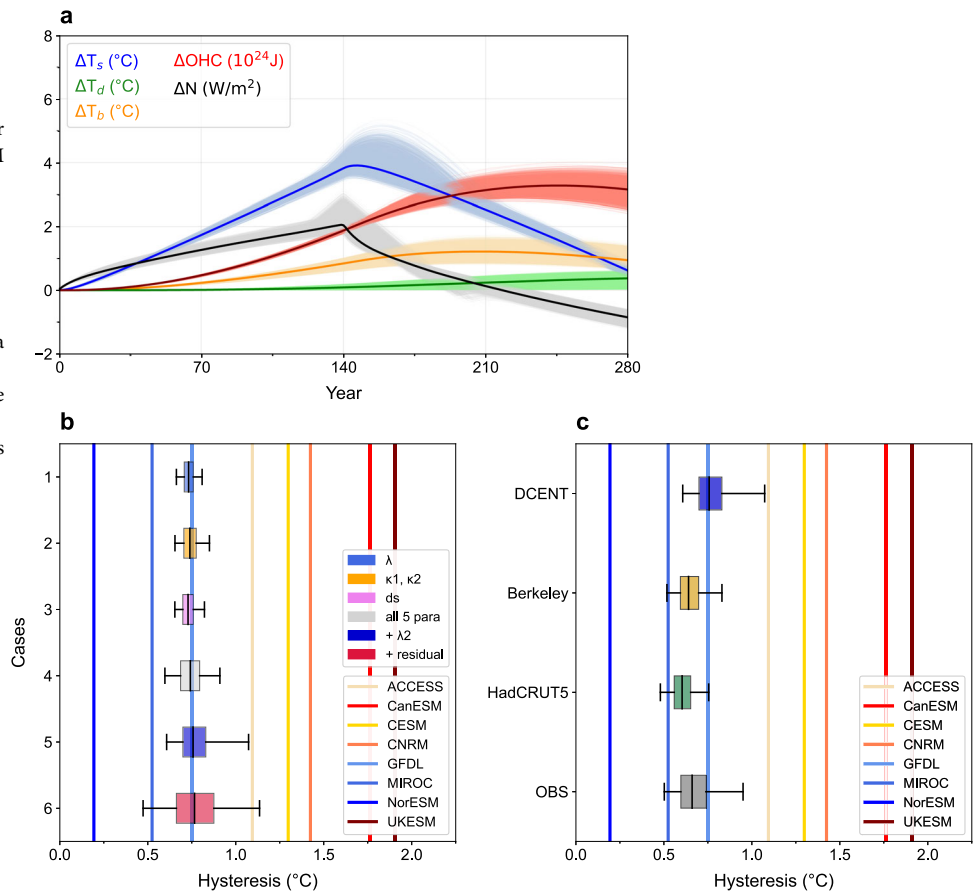
Constraining GMST hysteresis using historical observations

With EBM3 established, we employ it to constrain the GMST evolution under the CDR scenario. To ensure consistency with the observed global energy budget, we first calibrate the EBM3 parameters using historical observations from the Dynamically Consistent Ensemble of Temperature (DCENT) surface temperature⁴⁸ and the Institute of Atmospheric Physics (IAP) ocean temperature⁴⁹ datasets (see “Methods” section; Fig. 2a) following the same Bayesian approach previously used to obtain the parameter distribution. This framework enables a process-based understanding of potential uncertainty arising from structural differences in EBM3 parameters and observational products, while providing an observationally constrained estimate of GMST hysteresis.

EBM3 with parameters fitted to observations has been shown to capture the historical warming trend in response to radiative forcing from 1850 to 2020 (Fig. 2b). The emulation is constrained by the best estimate of the EBM3 parameters (Supplementary Table 4): λ has a median of 1.44 W m^{−2} °C^{−1} [1.33, 1.55], the ocean heat uptake coefficients (κ_1 and κ_2) have 0.86 W m^{−2} °C^{−1} [0.78, 0.96] and 0.56 W m^{−2} °C^{−1} [0.11, 1.35], respectively, and 104 m [80, 135] for the effective depth of the surface (d_s). Uncertainties are reported as the 95% confidence interval (c.i.) if not otherwise specified. This observationally calibrated EBM3 is then used to produce a 10,000-member ensemble of global temperature response under the CDR scenario (see “Methods” section). Notably, historical warming, which is less than 2 °C level, is too low to reach any of the critical temperature thresholds associated with changing λ , as diagnosed in any ESM simulations (Supplementary Fig. 2). As a result, EBM3 fit to observations does not have the non-linear changes in λ embedded.

To represent the potential non-linear behaviour of λ , evident in some ESMs but not constrained by historical observations, we randomly sample the estimated changes in λ ($\Delta\lambda$) and the critical temperature threshold from the Bayesian posteriors pooled across all eight ESMs for each member of our EBM3 emulation ensemble (Supplementary Table 2). Notably, three EBM3 (ACCESS-ESM1-5, CESM2, and NorESM2-LM) show a distinct change in λ , while the change in the other models remains below 0.1 W m^{−2} °C^{−1} (Supplementary Fig. 2). A negative $\Delta\lambda$ increases temperature sensitivity to

Fig. 3 | Global warming hysteresis emulated by EBM3 calibrated using observational estimates and its structural uncertainty. **a** Time series of N , T_s , T_d , T_b , and OHC anomalies from EBM3 emulations calibrated using the DCENT and IAP under the CDR scenario. Thin lines show individual EBM ensemble members, and thick lines denote their ensemble median. **b** Global warming hysteresis projection from CMIP6 ESM simulations (vertical lines) and EBM3 emulations calibrated using the DCENT and IAP datasets (box plot). Box plot indicates median (centre line), interquartile range (box), and 95% c.i. (whiskers). Each box represents a sensitivity test on EBM3 parameters, with all other parameters except those listed in the legend set to the ensemble median (see “Methods” section for a detailed description of the sensitivity test). **c** Same as (b), but for projections from EBM3 emulations calibrated using individual observational surface temperature estimates—DCENT, Berkeley Earth, and HadCRUT5—as well as their combined ensemble (OBS) to represent observational uncertainty.



forcing after the threshold is exceeded, potentially further amplifying GMST hysteresis (as discussed below).

The resulting median estimate of EBM3 emulation shows that GMST warms 3.8°C at the peak of CO_2 forcing, recovering only about 83% when the CO_2 forcing returns to its initial level (Fig. 3a). It provides the observation-constrained estimate of GMST hysteresis, yielding a median of 0.76°C [0.61 to 1.07°C] (Fig. 3b). Notably, ESM-based projections of GMST hysteresis diverge widely (Supplementary Fig. 1b), due in part to substantial structural differences across models²³ and the limited availability of computationally expensive simulation sets²⁶. In this regard, our estimate based on a physically improved EBM3 emulation calibrated with historical observations markedly reduces the inter-model uncertainty in ESM-based projections from CMIP6 CDRMIP, which range from 0.19 to 1.91°C (Fig. 3b).

Our framework also allows for comprehensive uncertainty quantification in projecting GMST hysteresis. This includes the parameter uncertainty from Bayesian inferences and structural uncertainty in the EBM3. We first address parameter uncertainty through sensitivity tests by perturbing the key parameter while holding the others at their ensemble median values (see “Methods” section). The uncertainties in GMST hysteresis, expressed as the 95% c.i. in $^\circ\text{C}$, are [0.66, 0.81°C], [0.65, 0.85°C], and [0.65, 0.82°C] when varying λ (case 1), κ_1 and κ_2 (case 2), and d_s (case 3), respectively (Fig. 3b). When all these parameters are allowed to interact (case 4), the 95% c.i. for GMST hysteresis increases to [0.60, 0.91°C]. These uncertainty estimates are inherent, as the Bayesian method infers the posterior parameter distribution in a probabilistic manner^{50,51}. As a result, the uncertainty arising from κ and d_s is comparable to that from λ , because λ is much better constrained by observations (Fig. 2a), even though it is a major factor in global warming projections by ESMs^{52,53}.

We next turn to structural uncertainty associated with the unconstrained changes in λ . Allowing λ to vary with further warming by randomly sampling $\Delta\lambda$ and the warming threshold derived from an ESM-informed

distribution (see “Methods” section) increases the GMST hysteresis uncertainty further to [0.61, 1.07°C] (case 5 in Fig. 3b), raising the upper bound relative to [0.60, 0.91°C] when $\Delta\lambda$ is neglected (case 4 in Fig. 3b). This increase is driven primarily by draws from NorESM2-LM, which exhibits the largest $\Delta\lambda$ of $-1.39\text{ W m}^{-2}\text{ }^\circ\text{C}^{-1}$ [-1.60 , -1.19]. Because $\Delta\lambda$ is less than $0.1\text{ W m}^{-2}\text{ }^\circ\text{C}^{-1}$ in five out of the eight ESMs considered, the lower bound and the median hysteresis estimate remain mostly unchanged. Note that we necessarily treat the available ESMs as an informative prior on account that $\Delta\lambda$ cannot be constrained by historical observations; the resulting uncertainty range is therefore conditional on this prior and could either overstate or understate the true structural uncertainty. Even so, the sensitivity of the upper tail underscores the value of explicitly accounting for $\Delta\lambda$ when emulating climate responses under the CDR scenario. Finally, to capture additional emulator uncertainty, we sample residual errors associated with reproducing GMST hysteresis in ESMs (case 6), which further broadens the range to the reported [0.47, 1.14°C] (Fig. 3b).

Another important source of uncertainty not yet accounted for is the existing discrepancies among historical observations. This is especially relevant for surface temperature estimates as DCENT suggests continuous warming since the 1850s, whereas other GMST products, including Berkeley Earth⁵⁴ and HadCRUT5 (Hadley Centre–Climatic Research Unit, v5)⁵⁵ estimates, all indicate a cooling over 1850–1910 (Supplementary Fig. 7b, d)⁴⁸. Given these observational differences, we repeat the same analysis using each of these other estimates (Supplementary Figs. 7 and 8). The resulting GMST hysteresis estimates based on Berkeley Earth and HadCRUT5 are 0.64°C [0.52, 0.83°C] and 0.60°C [0.48, 0.75°C], respectively (Fig. 3c). While our goal here is not to assess the reliability of individual datasets, this result points to a plausible and concerning outcome: GMST hysteresis constrained by more recent temperature estimates that better account for data biases^{48,56}, i.e., DCENT, could be larger than those relying on earlier datasets. Hence, substantial uncertainty can arise from discrepancies among historical

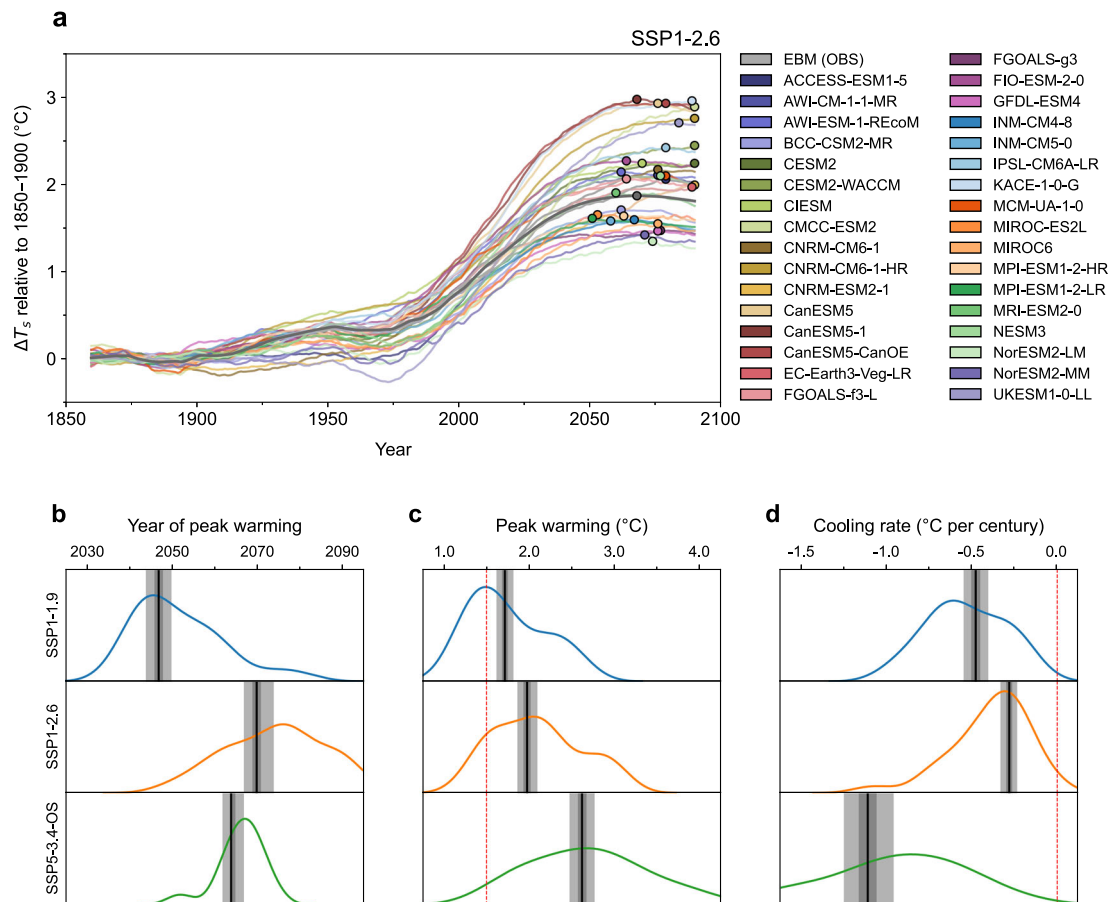


Fig. 4 | Future global warming emulated by EBM3 calibrated using observational estimates. **a** Time series of the 21-year running mean of T_s change relative to 1850–1899 from CMIP6 ESM simulations and EBM3 emulations combining results from calibrations with DCENT, Berkeley Earth, and HadCRUT5 (OBS) under the SSP1-2.6 scenario, with markers indicating the peak warming. Grey thin lines show individual EBM ensemble members, and thick lines denote their ensemble median. **b–d** Probability density distributions of CMIP6 ESM projections for the year of peak warming (**b**), the magnitude of peak warming (**c**), and the post-peak cooling rate (**d**)

(see “Methods” section for calculation details) under different SSP scenarios—SSP1-1.9 (blue), SSP1-2.6 (orange), and SSP5-3.4-OS (green). Vertical bands represent EBM3 projections combining results from calibrations with DCENT, Berkeley Earth, and HadCRUT5, with the centre line indicating the median, with dark and light grey shading representing the interquartile range and 95% c.i., respectively. Vertical red dashed lines in (**b**, **c**) indicate the 1.5 °C warming level and the 0 °C per century trend, respectively.

temperature estimates, and pooling the three with equal weight yields our final estimate of 0.66 °C [0.50, 0.95] (Fig. 3c).

We further assess the robustness of our EBM3 constraints as a function of the window length used to define hysteresis (Supplementary Fig. 9). With increasing window length, EBM3 provides tighter constraints on GMST hysteresis, while the inter-model spread across ESMs increases. Specifically, our observationally constrained GMST hysteresis reduces the uncertainty range by up to 74% relative to the ESM spread when hysteresis is defined over the full ramp-up (1–139 years) and ramp-down (141–279 years) periods. Taken together, these results demonstrate that our framework constrains uncertainties in ESM-based projections that rely on limited simulation sets and enables more robust constraints through improved representation of time-varying climate feedbacks and higher observational fidelity.

Implications for GMST trajectories under mitigation scenarios

Our framework is also applicable to emulate GMST trajectories under shared socioeconomic pathways (SSP) scenarios that include phases of both positive and negative CO₂ emissions, such as the SSP1-1.9, SSP1-2.6, and SSP5-3.4-OS, which reach net-zero CO₂ emissions in the 2050s, 2070s, and 2060s, respectively (Supplementary Fig. 10). Although most ESM simulations suggest a temporary overshoot of temperature above the 1.5 °C limit before temperature declines (Fig. 4), the timing and magnitude of the peak, as well as the pace of post-peak cooling rate, remain highly uncertain across

models. Taking the SSP1-2.6 scenario as an example, the 33 available ESMs project peak warming to range from 1.34 to 2.97 °C between 2051 and 2090 and a post-peak cooling rate from −1.08 to −0.05 °C per century (Fig. 4; see “Methods” section for calculation details). In contrast, our EBM3 emulation gives much tighter projections, expecting peak warming of 1.86 °C [1.68, 2.06] around 2068 [2066, 2073], and the post-peak cooling at the rate of −0.28 °C per century [−0.34 to −0.22].

Similarly, our EBM3 projects much tighter values for the SSP1-1.9 and SSP5-3.4-OS scenarios (Supplementary Fig. 11). Although we have accounted for both EBM3 parameter uncertainties and structural uncertainties in observational datasets (Supplementary Fig. 12), the estimates of peak warming are at least 75% narrower than the spread across ESMs, partly because climate feedback and sensitivity are much better constrained in our analyses. It is also worth noting that, unlike ESM projections that suggest a 38% (5 out of 13 available ESMs) chance of limiting warming under 1.5 °C with drastic emission reductions, our projection indicates a peak warming of 1.62 °C [1.47, 1.79] under the SSP1-1.9 scenario (Fig. 4b), indicating the necessity of planning for higher warming threshold targets.

Discussion

Current ESMs entail substantial uncertainty in projecting GMST recovery due to the limited simulation sets conducted to date and their structural differences^{23,26}. To tackle this limitation, we present a data-driven emulator framework by establishing an EBM3 that accounts for the non-linear

temperature response to CDR, with its parameter calibration using historical observations. This approach provides an observation-constrained estimate of GMST hysteresis and overshoot, which is notably narrower than ESM projections. Moreover, the structural uncertainties in the EBM3 emulation point to the need for addressing time-varying climate feedback in climate models and refining the fidelity of historical observations to better constrain GMST projections.

This study directly targets both an idealized CDR and overshoot scenarios, leveraging the EBM3 framework to emulate GMST trajectories during phases of positive and negative CO₂ emissions. More broadly, the framework offers a computationally efficient way to explore how uncertainties in GMST response under mitigation might behave across other forcing pathways—such as varying CO₂ ramp rates and geoengineering scenarios—that are impractical to sample comprehensively with ESM ensembles.

Although GMST is essential, the patterns of warming, such as the interhemispheric temperature difference and delayed Southern Ocean warming, are equally important, especially for understanding changes in the global hydrological cycle and regional climate risks^{57,58}. Changes in temperature patterns, however, remain even more uncertain across ESMs^{59,60}. In future undertakings, our framework could be extended to resolve latitudinal variations and constrained with gridded observational datasets, thereby improving regional climate projections and enhancing their relevance for climate risk assessment.

Methods

CMIP6 simulations

As for the CDR scenario, we utilized four experiments from 8 CMIP6 ESMs (ACCESS-ESM1-5, CanESM5, CESM2, CNRM-ESM2-1, GFDL-ESM4, MIROC-ES2L, NorESM2-LM, and UKESM1-0-LL)³⁶. First, we used the abrupt quadrupling of CO₂ forcing simulation (abrupt-4xCO₂) branched from each model's pre-industrial control simulation (piControl); for CESM2, the abrupt-2xCO₂ experiment was also used (see Supplementary Fig. 4). The abrupt-4xCO₂ experiment was primarily used to calibrate the EBM parameters, which were then applied to emulate the GMST trajectories under the CDR scenario. We also used the CDR-reversibility scenario from the CO₂ ramp-up (1pctCO₂) and ramp-down (1pctCO₂-cdr) experiment, which is integrated from the piControl simulation. In this experimental setup, atmospheric CO₂ forcing was prescribed to increase by 1% per year, reaching four times after 140 years from its initial pre-industrial level of 284.7 ppm, followed by a symmetric 1% per year decline until it returned to 284.7 ppm. In the CDR scenario, the Δ symbol denotes changes relative to the piControl simulation.

For the SSP scenarios, we employed 14 ESMs for SSP1-1.9, 33 ESMs for SSP1-2.6, and 13 ESMs for SSP5-3.4-OS from CMIP6³⁶. The corresponding model lists are shown in the legends of Supplementary Fig. 11a for SSP1-1.9, Fig. 4a for SSP1-2.6, and Supplementary Fig. 11b for SSP5-3.4-OS. We analysed climate projections for 2015–2100, which are branched from the historical simulation for 1850–2014. The SSP1-1.9 scenario largely reflects the 1.5 °C target under the Paris Agreement, reaching net-zero CO₂ emissions in the 2050s; SSP1-2.6 is a low-emissions pathway reaching net-zero CO₂ emissions in the 2070s, whereas SSP5-3.4-OS initially follows high emissions until 2040 before rapidly declining to net-zero CO₂ emissions in the 2060s (Supplementary Fig. 10).

Hysteresis definition sensitivity

We estimated GMST hysteresis as a measure of the long-term effects of CDR following the CO₂ peak, calculated as the temperature difference between the ramp-up (1–139 years) and ramp-down (141–279 years) periods at the same atmospheric CO₂ level under the standard CDRMIP 1% CO₂ ramp rate CDR-reversibility protocol⁹. This reflects the fact that hysteresis requires time to fully develop under a given CDR scenario, as delayed ocean heat uptake spreads warming over time and shapes post-CDR warming trajectories on multi-decadal to centennial timescales⁴⁴.

While hysteresis is known to depend on both the CO₂ ramp rate and window length^{20,23}, the CO₂ ramp rate is a property of the experimental

setup rather than a tunable analysis choice in our study relying on the 1% CO₂ ramp-up and ramp-down experiment. Accordingly, we assess the robustness of our main conclusions with respect to the choice of averaging window length. This includes an evaluation in the RMSE of GMST hysteresis between EBM2 and EBM3 relative to ESMs, which shows that EBM3 exhibits systematically lower RMSE as the window length increases to longer than 70 years (Supplementary Fig. 1c). Moreover, our observational constraints yield tighter hysteresis estimates, even as the spread of ESM-based projections increases with window length (Supplementary Fig. 9). These results indicate that our conclusions are robust to the choice of averaging window length, particularly for the longer window lengths of interest in this study.

EBM emulation

We introduce a simple EBM emulator with two-layer (EBM2) and three-layer (EBM3) configurations to project global temperature trajectories in response to changes in radiative forcing^{33,45}. The EBM2 formulation is as follows (see Fig. 1a for a schematic):

$$\begin{aligned} c_p \rho d_s \frac{dT_s}{dt} &= -\lambda T_s + F - \kappa_1 (T_s - T_d) \\ c_p \rho d_d \frac{dT_d}{dt} &= \kappa_1 (T_s - T_d) \end{aligned} \quad (1)$$

where T_s and T_d denote the temperature in the surface and deep ocean layers, respectively, F is the radiative forcing, λ is the climate feedback parameter, and κ_1 is the heat transfer coefficient between the surface and deep ocean layers. The term $c_p = 4180 \text{ J kg}^{-1} \text{ °C}^{-1}$ is the heat capacity of seawater, and $\rho = 1030 \text{ kg m}^{-3}$ is the seawater density. These two terms are multiplied by the effective depth of the surface (d_s) and deep layers (d_d) to represent the heat capacity of the corresponding ocean layers. The EBM2 parameters are estimated from the abrupt-4xCO₂ simulation using Bayesian inferences (Supplementary Fig. 2), and the posterior distributions for relevant parameters are reported with the median and 2.5–97.5% range (Supplementary Table 1).

We also employ the EBM3 to effectively emulate the global temperature response under the CDR scenario, which is given by (see Fig. 1c for a schematic):

$$\begin{aligned} c_p \rho d_s \frac{dT_s}{dt} &= -\lambda T_s + F - \kappa_1 (T_s - T_d) \\ c_p \rho d_d \frac{dT_d}{dt} &= \kappa_1 (T_s - T_d) - \kappa_2 (T_d - T_b) \\ c_p \rho d_b \frac{dT_b}{dt} &= \kappa_2 (T_d - T_b) \end{aligned} \quad (2)$$

where T_s , T_d , and T_b denote the temperature in the surface, deep, and bottom ocean layers, respectively. F and λ indicate radiative forcing and climate feedback parameter, respectively. c_p and ρ are the same constraint values as in EBM2, and these two terms are multiplied by the effective depth of the surface (d_s), deep (d_d), and bottom (d_b) ocean layers to represent the heat capacity of the corresponding ocean layers. κ_1 and κ_2 represent the heat transfer coefficients between the surface and deep ocean layers, and between the deep and bottom ocean layers, respectively. To better account for the non-linear temperature response in our EBM3 emulator, we allow λ to vary depending on critical temperature thresholds (T_{thres}) given by

$$\lambda(T_s) = \begin{cases} \lambda_1 & \text{for } T_s < T_{\text{thres}} \\ \lambda_2 & \text{for } T_s \geq T_{\text{thres}} \end{cases} \quad (3)$$

where λ_1 and λ_2 denote climate feedback parameters below and above T_{thres} , respectively. All these EBM3 parameters are jointly estimated from the

abrupt-4×CO₂ simulation using Bayesian inference by iteratively updating the parameters to minimize the mismatch between EBM-emulated responses and the corresponding ESM outputs (see the “Methods” section on Bayesian inference for details). As a result, a total of 10,000 posterior samples are used as a probabilistic representation of the EBM3 parameters, and the posterior distributions are reported with the median and 2.5–97.5% range (Supplementary Table 2).

Based on the EBM parameters estimated from the abrupt-4×CO₂ simulation, we then emulate the global temperature trajectories by prescribing the radiative forcing from ref. 61 for the historical period and from IPCC AR6 best estimate⁶² for the SSP scenarios. To ensure consistency between the historical and SSP scenario radiative forcing datasets, the SSP forcings were offset to match the historical values over the overlapping years 2010–2020 (Supplementary Fig. 13).

It is important to note that the non-linear changes in λ of EBM3 were embedded for scenarios exceeding the critical temperature threshold, which needs to be at least as warm as the lowest threshold estimate of 3 °C [2.73, 3.40] from GFDL-ESM4 (Supplementary Table 2). In the EBM3 emulation, this threshold is exceeded only under the CDR scenario. Therefore, we randomly sampled the λ_2 and T_{thres} from each CMIP6 ESM, and then allowed λ to vary depending on whether T_s exceeds T_{thres} under the CDR scenario (Supplementary Fig. 6).

Bayesian inference

We applied the Bayesian inference to condition the EBM parameters. The Bayesian modelling platform PyMC⁶³ is used to obtain the joint posterior distribution via the Hamiltonian Markov Chain Monte Carlo (HMC) sampler and a No-U-Turn sampler⁶⁴. Normal priors, $N(1, 0.5)$, that are truncated to be greater than zero are prescribed for λ and κ , and $N(100, 50)$ for d_s . The posterior distribution for the EBM parameters was inferred from 2000 draws across five chains, each sampled after an initial burn-in of 10,000 tuning steps. The Bayesian inference iteratively updates the EBM parameters to maximize the likelihood of the EBM-emulated responses given the corresponding ESM outputs, yielding a total of 10,000 posterior samples for each parameter in EBM2 (Supplementary Table 1) and EBM3 (Supplementary Table 2). In EBM3, the inference explicitly includes the state-dependent parameters λ_1 , λ_2 , and T_{thres} , thereby allowing the climate feedback parameter to vary across different warming states.

Specifically, we used this Bayesian framework to condition EBM parameters on each ESM simulation from the first 150 years of the abrupt-4×CO₂ simulation (Supplementary Fig. 2) to emulate the global temperature trajectories under the CDR-reversibility scenario (Supplementary Fig. 3). We also fitted the EBM3 parameters to instrumental records of Earth surface and ocean interior temperatures (Fig. 2 and Supplementary Fig. 7, see Historical temperature observations in the method section for more details). By calibrating the EBM3 parameters with historical observations, we provide constraints on the global temperature trajectories under the CDR-reversibility (Fig. 3) and the SSP1-1.9, SSP1-2.6, and SSP5-3.4-OS scenarios (Fig. 4 and Supplementary Fig. 11).

Historical temperature observations

To calibrate EBM parameters, we employ the historical observations from the DCENT surface temperature⁵⁴ and the IAP ocean temperature⁵⁵. The DCENT surface temperature estimate has employed several state-of-the-art data homogenization techniques to almost fully remove data artefacts, especially for early biases in sea surface temperature (SST) measurements⁴⁸. SST corrections in DCENT are also better in line with evolutions of the amplitude of diurnal cycles, and its SST estimates are more consistent with proxy SSTs based on tropical corals and expected temperature evolution from radiative forcing⁶⁵. The IAP product gives reliable ocean temperature estimates after accounting for biases in expendable bathythermograph (XBT) and mechanical bathythermograph (MBT) measurements via comparisons with nearby conductivity-temperature-depth (CTD) records^{66,67}.

To account for observational uncertainty, we also use the Berkeley Earth temperature⁵⁴, and HadCRUT5⁵⁵ (Hadley Centre–Climatic Research Unit, v5) surface temperature estimates. While DCENT suggests continuous warming since the 1850s, other GMST products, Berkeley Earth⁵⁷ and HadCRUT5⁵⁸, indicate cooling during 1850–1910 (Supplementary Fig. 7b, d). Because DCENT only reports temperatures at locations directly sampled, we generate globally complete fields by ordinary kriging using the GloMar Gridding System (ref. 68.) before calculating the GMST series. We exclude GMST estimates from NOAA Global Temperature V6⁶⁹ because its SST component, i.e., ERSST5⁷⁰, includes late 19th century adjustments that can leave a residual cooling bias in the 1850–1900 trend, potentially understating centennial-scale warming and thus biasing inferred climate sensitivity and projections⁵⁶.

Each GMST series is paired with the same 1960–2020 IAP ocean temperature estimate⁴⁹. Likelihoods are evaluated over the available period for each observable, i.e., 1850–2020 for GMST and 1960–2020 for OHC. We fit the model to each GMST-OHC pair under identical priors, drawing 10,000 posterior samples per fit. As a result, we emulate a 10,000-member ensemble of global temperature responses for each GMST-OHC pair from DCENT, Berkeley Earth, and HadCRUT5 under the CDR-reversibility and the SSP1-1.9, SSP1-2.6, and SSP5-3.4-OS scenarios, and pool the 30,000 draws together with equal weight to obtain the final observationally constrained posterior.

EBM parameter sensitivity test

In Fig. 3b, we present a series of sensitivity tests to assess the impact of EBM3 parameter uncertainty on GMST hysteresis projection. Six cases were designed by perturbing key EBM3 parameters while fixing the others at their ensemble median values. In Case 1, λ was varied. Case 2 involved κ_1 , and κ_2 , while in Case 3, changes are applied only to d_s . In Case 4, all aforementioned parameters were allowed to vary. Case 5 introduced a non-linear dependence of λ on warming threshold, allowing λ to change when the warming exceeds certain thresholds based on the ESM-based estimate (Supplementary Fig. 2). Case 6 accounted for structural uncertainty by sampling from the residual distribution of the EBM3.

Peak warming and post-peak cooling rate

To estimate the key metrics that will influence post-net-zero climate outcomes—namely, the timing and magnitude of peak warming, and the rate of post-peak cooling—we first smooth the GMST anomaly relative to the 1850–1900 baseline using a 21-year running average. We identify peak warming as the year in which the smoothed GMST value reaches its maximum. Post-peak cooling rates are then calculated using a linear regression to the smoothed GMST from the peak warming year to 2100. For ESMs where peak warming occurs after 2090, the relatively short post-peak period precludes reliable estimation of cooling rates, and these simulations are therefore excluded from the analysis.

Data availability

The model data used in this study are freely available from the CMIP6 archives at <https://esgf-node.llnl.gov/projects/cmip6>. Observational temperature estimates are publicly available as follows: DCENT (<https://doi.org/10.7910/DVN/NU4UGW>), HadCRUT5.0.2.0 200-member ensemble and uncertainty estimates (<https://www.metoffice.gov.uk/hadobs/hadcrut5>), Berkeley Earth temperature (<https://berkeleyearth.org/data/>) for surface temperature, and IAP for ocean temperature datasets. The source data used to generate the figures are available from ref. 71.

Code availability

The code for the analysis and figure generation used in this study is available from ref. 71.

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Author contributions

S.-Y.S. and D.C. conducted the analysis and generated the figures. S.-Y.S., D.C., and S.-W.Y. conceived the study, discussed the results, and revised the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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